

PlumJob recruitment support system

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Software Requirements Specification

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Krakow 16.06.2025

Table of Contents

1. Introduction

- 1.1 Purpose of the Document
- 1.2 Product Scope

2. General Description

- 2.1 Target Audience
- 2.2 Operating Environment
- 2.3 Constraints
- 2.4 Technologies Used

3. Technologies Used

- 3.1 Java
- 3.2 Spring boot
- 3.3 MySQL
- 3.4 JavaScript
- 3.5 Vaadin
- 3.6 Amazon Web Services
- 3.7 Figma and Canva

4. UI description

5. Functional Requirements

6. Non-Functional Requirements

1 Introduction

1.1 Purpose of the Document

This document describes the requirements and constraints that the project must adhere to. It outlines its functionalities and the conditions that the system must fulfill.

1.2 Product Scope

A web application designed to support the job search process. The system enables users to add job applications they are interested in, including the job position, company name, start date, and a description. The user can view the history of both ongoing and completed applications using a few simple search filters. Based on this history, a graph can be generated to reflect the user's data. In both cases, it is possible to update the current status. The system also allows the generation of resumes using provided templates. The user can read articles aimed at facilitating future recruitment efforts. Users are identified through a unique username and password.

2 General Description

2.1 Target Audience

The project is intended for individuals currently seeking employment who want to better manage the entire process, as well as for employers looking for a platform to advertise themselves to potential candidates.

2.2 Operating Environment

The application is designed for the Microsoft Windows operating system, particularly Windows 10. It requires Java Runtime Environment, a web browser, an internet connection, and a MySQL database to run.

2.3 Constraints

Each user must have a unique login.

The application requires an internet connection to function.

The CV builder is limited to predefined templates.

Calendar is limited to selecting only whole hours.

3 Technologies Used

3.1 Java

Java, along with the Spring Boot framework mentioned below, is responsible for virtually the entire backend. It was chosen

because the entire team had prior experience with it during the *Programming Methods* course.

3.2 Spring Boot

Also responsible for the backend. It was chosen because it is an industry standard and is compatible with Java.

3.3 MySQL

Used for the database. It was chosen due to being an industry standard and because the entire team is familiar with this technology.

3.4 JavaScript

Used to create the CV builder and flow diagram. It was chosen because it allows for easier implementation of these components.

3.5 Vaadin

A Java framework used for frontend development. It was chosen due to its UI compatibility with Spring Boot and the availability of good learning materials. Additionally, it allows both frontend and backend development within a single project.

3.6 Amazon Web Services

A service used for hosting the database. It was chosen because it offers security and its Tier 1 is free while still providing all the services we need.

3.7 Figma and Canva

Software used for designing the UI.

4 UI description

UI description can be found in a separate file named

5 Functional Requirements

- Users can register and log in to their accounts. Their data is securely stored in a connected database.
- Users can save and retrieve their job applications. Applications are displayed in two tabs: active applications and archive.
- Users can search through applications using filters. They can view both active and archived applications, display only selected application statuses, and sort results in ascending or descending order by application name, date added, or end date.
- The system selects and displays advertisements with job offers relevant to the user.
- Users can browse recruitment tips and articles in a separate "tips" tab.

- The system allows the user to generate a CV using one of several templates in a dedicated tab. After selecting the desired position and filling in the required information, the system saves the file in PDF format.
- The system can generate a graph representing the user's recruitment history based on the database information.
- The system displays the number of jobs found using the application based on user declarations.
- Users can save important dates in a calendar and they will be reminded of them by notification.
- System recommends potential job position based on user declared tags.

6 Non-Functional Requirements

- Passwords are being encrypted.
- The database is failure-resistant and stores backups thanks to security mechanisms provided by Amazon Web Services.
- The system is open to expansion through the addition of new CV templates, recruitment support articles, and new advertisements for monetization purposes.